

INF201 - Deliverable 1

For this deliverable you will be working with weather data from the The Norwegian Meteorological Institute.

Getting started

1. Create a user at frost.met.no

This will give you a client-ID you can use to access data from The Norwegian Meteorological Institute (MET).

A few things to keep in mind about this ID:

- The ID is linked to you, and should therefore only be used by you
- IDs like this should not be submitted to your GitHub repository, and should therefore not be part of your submitted code/notebooks.
- We recommend storing the ID in a file that is not part of your Git repository, and to read from that file when you request data

2. Look through this example from MET about [how to work with Frost using Python](#)

- Note that this example code is somewhat outdated, and you might need to re-write parts of it if you use it in your work.
- Remember to reference if you use this in your assignment.

Get data from Ås

- Explore the [available sources for frost.met.no](#)
 - Paste your client-id in the top right corner
 - Here you can explore what data is available and find out how to query it
 - If you write Ås in the name field, you will get information about our local weather station
 - Here you will see that the ID for our station is "SN17850"
 - [More information about weather data from NMBU here](#)
- Explore the [available types of weather data here](#)

Task 1: Temperature

- First request the daily mean air temperture for all of 2025.
 - The specifications you will need for this is:

```
endpoint = 'https://frost.met.no/observations/v0.jsonld'
parameters = {
    'sources': 'SN17850',
    'elements': 'mean(air_temperature P1D)',
    'referencetime': '2025-01-01/2025-12-31',
}
```
 - Otherwise you can follow the example from frost.met.no on how to get the data when using Python

- Make a nicely formatted plot of this air temperature, where time is on the x-axis and temperature is on the y-axis
- Create a nicely formatted summary table containing mean, median, min and max temperature for this year, and display it with `print()`

Task 2: Precipitation

- Request data for precipitation for the same time period, specifically measurements of precipitation over 24 hours. This is coded as `sum(precipitation_amount P1D)` in the database
- Create a new plot that shows both the temperature from task 1 and the precipitation in the same plot
 - Think through how to display both values in a way that is useful and looks good
 - Consider which type of plot works best for each measurement (scatter, line, bar, pie chart, ...?)

Task 3: Historical temperature data

- Now you want to compare the 2025 temperatures to temperatures measured 100 years earlier.
- Get these temperatures from the database.
- Plot the new and the old temperatures in the same plot, still nicely formatted
- Generate a summary table as for task 1, where you display data from both the new and the old temperatures.

Task 4: Heatwaves

- The next task is to look for heatwaves in Ås in 2025
- In Norway a heatwave is defined as a consecutive period of five or more day where the *max* temperature is ≥ 27 degrees Celcius.
 - Note that you need max temperature here, not mean
 - Max temperature is coded as `max(air_temperature P1D)`
 - [met.no article about heatwaves](#) (only in Norwegian)
- Write a function that count the number of heatwaves in a year.
 - The function should write a summary to a yaml file, formatted as the example file
 - Write the function so that it will work with any station name/id and year

Task 5: Moving average temperature

- Write a function that calculates the 7-day moving average for temperature
- Use this function on the 2025 mean temperature you collected in task 1
- Make a plot showing the moving average as a line and the daily temperature as points

Hand-in information

- This should be uploaded to Canvas by the deadline:
 - The Jupyter Notebook file containing the code, the results and/or plots generated by the code, discussion of the results in Markdown and a link to the GitHub repository you have used.
 - A pdf version of this notebook (see lecture 5)
 - A discussion of GenAI use, for example a summary of the chat log as you have done earlier
 - A short (5 mins) video discussing your code and results
 - Note that you both have to contribute to the video
 - Either by recording it together
 - Or recording parts separately and editing it together